ICNT86 Series Touch Control Chip Application Manual

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1. Overview

The ICNT86 series is a family of capacitive touch detection chips with a built-in MCU. They support multi-touch functionality (up to 10 touch points) and can form capacitive touch panel modules (CTPMs) when combined with capacitive touch screens. This document describes how to use ICNT86 series chips in systems and how the host communicates with them to realize multi-touch.

1.1 Function Overview

• Real multi-touch: Supports 5–10 points. • Supports DITO, SITO, OGS, and full ITO structures. • Strong anti-interference capability with built-in RF/LCD noise suppression and auto frequency hopping. • Supports passive stylus, glove touch, and gesture in standby mode. • Reporting rate: 80Hz (typical), up to 200Hz. • Power: 2.8V–3.3V; I/O voltage 1.8V or VCC (configurable). • Applications: smartphones, tablets, portable media players, GPS, cameras, etc. • I2C interface up to 400kbps. • Automatic calibration.

1.2 Product Selection

Models include ICNT8636, ICNT8656, ICNT8672, ICNT8682, ICNT8692, ICNT8698, etc., covering phones and tablets from 4.0” to 12.1”.

2. Function Structure and Working Modes

The ICNT86 integrates a 32-bit MCU, DSP, SRAM, Flash, and AFE (Analog Front End). The MCU/DSP processes touch data and coordinates control. AFE handles analog input and converts it to digital.

Operating Modes:

1. Normal Mode – Full scan operation. 2. Monitor Mode – Low-frequency scan for touch detection. 3. Hibernate Mode – Ultra-low power, waits for wakeup. 4. Gesture Mode – Detects standby gestures to wake system.

Automatic Frequency Hopping

The chip monitors noise across frequency bands and automatically switches to a cleaner band if interference exceeds a threshold.

3. CTPM Application

CTPM (Capacitive Touch Panel Module) includes the touch screen and ICNT86 chip. It connects to the host via I2C, interrupt (INT), reset (RST), and wakeup (WAKE) pins.

I2C Interface

Standard I2C communication, 7-bit address 0x48. Read/write controlled by LSB bit (0=write, 1=read). Supports up to 400kbps. Timing and sequence diagrams are provided in the original document.

Registers

Registers divided into: • Header area (0x0000–0x001F) • Coordinate info (0x1000–0x1FFF) • Configuration area (0x8000–0x8FFF) Includes touch coordinates, gesture IDs, system flags, etc.

4. Host Platform and Programming

Host must have I2C, INT, RST, and WAKE support. Power: 2.6–3.6V (recommended 2.8–3.3V).

Sleep and Wakeup

Sleep command (0x04 0x02) puts IC to low power. Wakeup via reset, GPIO, I2C, or gesture mode.

Interrupt Modes

Edge-triggered (rising/falling) or level-triggered (high/low).

Software Upgrade

Two methods: via programming tool or host OS (ADB/APK).

5. Important Statement

All technical information may be replaced by newer versions. Chipone makes no warranty for suitability or performance. Do not use in life-support systems without written approval.